CN TA-2

TCP HERO — THE MISSING PACKET

Detailed Project Brief

Roll Numbers	65 67 72
Project Type	Story Composition
Subject Area	Computer Networks (CN)
Technologies	HTML, CSS and JavaScript

1. Project Overview

TCP Hero — The Missing Packet is a comic-style educational website designed to explain how the Transmission Control Protocol (TCP) supports reliable and ordered delivery of data between applications. It presents a familiar example: a user sends a photo to a friend, the information is divided into smaller pieces, one piece goes missing during transmission, and the system must recover the missing data.

The project combines a visual story with simple interactive demonstrations and a short quiz. Rather than presenting networking concepts only as definitions, it shows the sender, receiver, data packets, acknowledgements, and retransmission as parts of a single sequence of events.

Purpose of the Project

- Make key TCP concepts easier to understand through a visual, story-based learning experience.
- Demonstrate how a receiver can detect that data is missing and how TCP recovery mechanisms help.
- Allow users to interact with a simplified network scene and test their understanding through a quiz.

The website is an educational model. Its animations illustrate the ideas but do not create a real TCP connection.

2. Problem Statement

Computer-network concepts can be difficult for beginners when they are introduced only through technical terminology. Data is transmitted across networks in segments, and the receiving application needs the data to be delivered in the correct order. Data may be delayed or lost while travelling through a network, so a transport protocol needs a way to track progress and recover when necessary.

The project addresses this learning challenge by providing a simple interactive explanation of TCP. Through the story of a missing packet, learners can follow how data is sent, how receipt is acknowledged, and how missing data may be retransmitted. The project is intended to support understanding, not to simulate every detail of TCP in a real network.

Objectives

- Explain TCP's role in providing a reliable, ordered byte stream between applications.
- Introduce the sender and receiver roles in a data transfer.
- Explain how sequence numbers track positions within a byte stream.
- Describe acknowledgements (ACKs) and how they report receipt progress.
- Illustrate packet/data loss and the idea of retransmission.
- Provide hands-on interaction through controls for sending data, returning an ACK, and triggering a loss-and-recovery scenario.
- Evaluate learning with a five-question multiple-choice quiz and score feedback.

Scope

The scope includes the website interface, comic panels, packet-motion animation, a small interactive lab, quiz questions, score calculation, and responsive styling for smaller screens. The site is intended to run locally in a modern web browser by opening the supplied index.html file. It does not require a live network connection for its educational animation.

Intended Users

The main audience is students learning Computer Networks, especially those who need an introductory explanation of TCP reliability. The comic format and short activities may also help general users who want a basic understanding of what happens when data does not arrive as expected.

3. Core Concepts Explained

Transmission Control Protocol (TCP)

TCP is a transport-layer protocol used for communication between applications. In this project, it is presented as a mechanism that helps deliver a byte stream reliably and in order. This means the receiving application can work with data in the intended sequence rather than having to manage all ordering and recovery tasks itself.

Sequence Numbers

TCP sequence numbers identify positions in the byte stream. They help the receiver determine which data has arrived and what data is expected next. If data is missing, the receiver can recognise that there is a gap rather than treating later data as if the earlier bytes had arrived.

Acknowledgements (ACKs)

An acknowledgement is feedback from the receiving side about data receipt progress. In the website's simplified explanation, an ACK commonly indicates the next byte expected. The sender can use acknowledgement information to track which data has been acknowledged and to manage ongoing delivery.

Retransmission

If TCP's recovery logic detects that data may have been lost, the sender can transmit the missing data again. Recovery can involve acknowledgements and timers, among other mechanisms. The website shows this as packet P2 disappearing and then being sent again. Actual TCP recovery depends on connection state, timers, congestion control, and implementation details.

Story Workflow

- **Step 1 — Message creation:** Nova wants to send a photo to a friend.
- **Step 2 — Data tracking:** The information is represented as smaller pieces, and sequence numbers help track byte positions.
- **Step 3 — Missing data:** P1 arrives, but P2 is missing while later data may already be seen.
- **Step 4 — Detecting progress:** The receiver tracks what it has and what it still expects; ACKs and timers help the sender determine when recovery is needed.
- **Step 5 — Retransmission:** The sender sends the missing data again.
- **Step 6 — Ordered delivery:** Once the required bytes are available, the receiving application can use the ordered stream.

This sequence is a simplified teaching example. It focuses on the main idea of reliable, ordered delivery rather than every possible TCP behaviour.

4. Main Features and Implementation

Comic-Style Story

The website uses a comic-inspired design to introduce the problem and its solution. It includes characters representing the sender, receiver, and a TCP helper, along with panels that show the message being prepared, data being split and tracked, the missing piece, and the recovery.

Interactive Packet Lab

The network control room provides four controls:

- **Send packets:** animates a data packet moving from the sender towards the receiver.
- **Send an ACK:** illustrates an acknowledgement travelling back to the sender.
- **Lose P2 & recover:** demonstrates a simplified loss event followed by retransmission of P2.
- **Reset scene:** returns the animation and status messages to their initial state.

The lab updates status text and explanatory messages as each action progresses. These controls are designed to reinforce the concepts visually; they do not send actual packets over the internet.

Quiz and Score

The final section contains five multiple-choice questions covering TCP's purpose, acknowledgements, sequence numbers, retransmission, and the difference between reliability, encryption, and speed. After a user selects an answer, the interface shows feedback and identifies the correct option. The score is updated, and a final message reflects the result.

Technologies Used

Technology	Role in the Project
HTML	Structures the page, story sections, buttons, quiz area and content.
CSS	Defines colours, typography, comic panels, layout, animation and responsive behaviour.
JavaScript	Controls packet animations, updates mission messages, handles quiz answers and calculates scores.

Responsive and Accessible Presentation

CSS media queries adapt the layout for smaller screens, including a single-column arrangement for panels. The website also includes visible keyboard-focus styling and a reduced-motion preference so animations can be limited for users who prefer less movement.

5. How to Run and Test the Project

Steps to Run

- Keep the supplied project files together in the same folder.
- Open the file named **index.html** in a modern web browser such as Chrome, Edge, or Firefox.
- Scroll through the comic story and read the explanations.
- In the interactive lab, try each control: send packets, send an ACK, lose P2 and recover, and reset the scene.
- Complete the five-question quiz and observe the answer feedback and final score.

The README states that Google Fonts are optional. If those fonts cannot load, the page uses available system fonts instead.

Suggested Functional Test Checklist

Test	Action	Expected Result
1. Open page	Open index.html in a browser.	The project title, story and navigation appear.
2. Send data	Click “Send packets”.	Packet animation moves towards the receiver and status changes.
3. Return ACK	Click “Send an ACK”.	ACK animation returns towards the sender.
4. Recover loss	Click “Lose P2 & recover”.	The scene shows loss, retransmission, and recovery messages.
5. Reset	Click “Reset scene”.	The scene returns to its initial state.
6. Quiz	Answer all five questions.	Feedback and score appear; a completion rank is shown.

Limitations

- The network animation is a simplified visual model, not a real TCP connection.
- The project focuses on basic reliability concepts and does not implement TCP networking, congestion control, or packet capture.
- The website depends on a modern browser for its interactive JavaScript and CSS behaviour.

Conclusion

TCP Hero — The Missing Packet presents TCP reliability in a visual and interactive format. By combining a comic story, a packet lab, concise explanations, and a quiz, it helps learners connect sequence numbers, acknowledgements, and retransmission with the goal of ordered data delivery. The project demonstrates how HTML, CSS, and JavaScript can be combined to build an engaging learning resource for Computer Networks.

Prepared from the supplied TCP Hero website source, README, and original project brief.